

Instance disambiguation across the cognition spectrum — a study design

Status: design / planned. No experiments have been run. This is a plan, shelved alongside the completed studies so the next phase is ready to build. It reuses the framework, harness architecture, and metrics of [REPORT.md](#); read that first for the semantic-model definition, the cognition spectrum, and the resolvability argument this design leans on. It is deliberately scoped to exercise **all** of the instance-level mechanisms the MAGIC paper conceives — not only co-reference from static evidence, but interrogation, the **virtual operation on the knowledge graph** checked against semantic invariants, confidence-held proposals corroborated by regeneration, and the **invariant** and **instance** reference variants — so that the demonstration matches the framework rather than a subset of it.

Summary

The schema-term studies ([REPORT.md](#), [REPORT_reference_anatomy.md](#)) reconcile the vocabulary of two models — which type denotes the same type as which. This design turns to the **instance level**: co-reference of *individuals* across two populated graphs — is this ROADM the same device as that te-node, this ODU2 service the same as that te-tunnel. It is a separate exercise that *follows* schema alignment (once the types are aligned, one can compare like with like) and runs on a different path: **entity resolution over the data**, from keys, attributes, and topology, rather than meaning-comprehension over the vocabulary.

Two things make it worth its own study, and both trace directly to the MAGIC paper. First, the cognition spectrum bites **harder** here: for the genuinely hard individuals — two structurally identical devices, or an individual with no shared key — the decisive evidence exists *only* by acting on the live system, so that whole class of evidence is gated by live cognition, and the residual-as-shortfall pattern becomes structural rather than incidental. Second, the paper's central verification mechanism is itself an instance-level act: a proposed correspondence is *exercised* by a **virtual operation on the knowledge graph** — provision a service through the candidate and confirm the resulting objects *in place* against a fixed set of **semantic invariants** (endpoint identity, connectivity, capacity, layer relationships, switching constraints, multiplexing structure), not by byte equality. Instance disambiguation is therefore both the cleanest test of the across-the-spectrum thesis and the natural home for the correctness-by-construction claim. This design covers the full mechanism set the paper names: co-reference from the static evidence hierarchy (key → attributes → topology), **interrogation** of a live side, the **virtual manipulation** as a first-class invariant-checked verification stage, **confidence-held proposals corroborated by regenerating native records**, the confirm-versus-propose split that governs how recall is read, and the **instance** and **invariant** reference variants of the reference family.

1. Motivation and relation to the schema studies

The main study established two things: that cognitive agents reconcile divergent models ad hoc, and that the *placement* of cognition governs what reconciliation can resolve — complete in principle between two fully-cognitive agents, with a growing residual (the shortfall from full cognition's reach) as cognition recedes. Those results are about the schema. Real interoperation also requires aligning the **individuals** the schemas classify, and that is a different problem with a different failure surface. This study asks the same governing question — how does the placement of cognition change what can be resolved, and what evidence carries the weight — but for instance co-reference, and it brings the paper's verification mechanism into the measurement rather than leaving it implicit.

2. The task

Given two populated graphs of one network and an **already-reconciled schema**, an agent must:

- decide which individual in graph A is the same real-world entity as which individual in B;
- hold each decision with a **confidence**, and corroborate it — by fit, by regenerating a native record, and, where cognition allows, by a virtual manipulation (§6);
- leave A-only and B-only individuals as **residual** (native gaps);
- **never merge two distinct individuals** (an instance false cognate);
- and, where identity cannot be settled or confirmed from the evidence available, refer the individual onward as residual rather than guess.

Scored against a validated **gold instance alignment**, derived from the case by construction so it cannot drift, exactly as the schema gold is. The gold additionally encodes, per individual and per candidate service, the **semantic invariants** a virtual manipulation is expected to confirm (§4), and flags the **experiment-only** subset (§4).

3. Why the cognition spectrum bites harder here

In schema reconciliation an inert side still exposes its whole structure and data statically, so the live agent can usually reconstruct a type's meaning from what is on the page; the virtual experiment is a confirmation, not the only route. Instance co-reference is different. Some individuals are distinguishable *only* by an active query or experiment against the live system — "what is the serial of the device at this port?", or *provision through candidate correspondence X and observe which real device lights up*. That class of evidence is available only under live cognition. Consequently:

- **both-cognitive** — resolution is complete in principle: any individual's identity can be confirmed by interrogation or a decisive virtual manipulation, so the residual can in principle reach zero.
- **one-inert** — the live side can still be interrogated and manipulated; the inert side is a static snapshot. Some individuals on the inert side that are separable only by experiment become unresolvable — unless a published reference supplies what the side cannot volunteer.
- **both-inert** — no probing or manipulation at all; a third party co-refers from static evidence only and can only *propose* candidates, leaving every experiment-only individual residual.

The shortfall from full cognition is thus not only present but **structural and unavoidable** for a definable subset of individuals — a sharper instance of the same law the schema study measured, and the reason the experiment-only subset is broken out in the gold and the metrics.

4. Case construction (the seeded hard instance case)

A purpose-built populated transport network — a fresh case, in the same domain and over the **already-reconciled TAPI/TEAS schema types** of the first case, so the "schema already reconciled" premise holds and continuity with the programme's setting is preserved, while the A-box topology is engineered to plant clean, *provable* traps. The individuals are deliberately booby-trapped so that different evidence types are each made to matter:

- **Merge targets** — the same real device under different local ids/names in the two graphs; must be merged. Separable by attributes or topology, not by name.
- **Instance false cognates** — distinct individuals that share a local name (two `R1`s in different sub-domains; two `svc-100`s) or an attribute value; must **not** be merged.
- **Structurally symmetric pairs** — two individuals with identical local topology, distinguishable only by a global key or by a virtual manipulation. These are the experiment-only cases, engineered so that static evidence genuinely underdetermines them.
- **Native gaps** — A-only and B-only individuals; correctly residual.
- **Keyless ambiguous** — individuals with no shared key and under-determining attributes; residual unless interrogation or manipulation is available.

Semantic invariants in the gold. Because the paper's verification is confirmation against invariants rather than byte equality, the case encodes, for each individual and for the services that can be provisioned over it, the invariant values a correct correspondence must preserve: **endpoint identity, connectivity, capacity, layer relationships, switching constraints, and multiplexing structure**. A virtual provision through a candidate co-reference (§6) is scored *confirmed* only if the resulting objects match these invariants in place, and *refuted* if any diverges — which is exactly how a wrongly-merged structurally-symmetric pair is caught. The gold records the correct co-reference pairs, the planted instance false cognates that must not be merged, the native gaps expected as residual, the per-correspondence invariant signature, and the subset flagged **experiment-only** (resolvable in principle solely by interrogation or manipulation), which lets us measure the shortfall directly as cognition recedes.

5. Evidence factors (what to ablate)

The instance analogue of the reference fields — the *static* evidence an agent may use to co-refer individuals before any interrogation or manipulation. These are the factors of the ablation:

- **Global key** — a shared identifier across both graphs; the instance analogue of the concept id anchor. It trivializes co-reference when present, so — heeding the id-slug lesson from the reference-anatomy study — keys are **opaque** (`k01`, ...), never leaked descriptors, and their presence/absence is a factor.
- **Local name/label** — the trap surface; where the instance false cognates live.
- **Attribute vector** — properties such as rate, coordinates, address, admin-state.
- **Relational/topological signature** — the individual's edges and neighbours; the hypothesised workhorse among static features, since an individual is largely defined by what it connects to (the A-box parallel to *definition* for concepts).
- **Provenance / temporal** — source and timestamp; optional, a secondary factor.

6. Live cognition, operationalized: interrogation and the virtual manipulation

The genuinely new machinery, and the crux of the study. Live cognition acts on the knowledge graph in two distinct ways, both of which the MAGIC paper uses and both of which the harness must provide through a single **oracle** on a live side, with every call counted:

1. **Interrogation (evidence-gathering)**. The agent asks an authoritative question about an individual on a live side — a single authoritative attribute, or the response to a candidate operation ("issue operation *o* and read how the system responds"). This *reconstructs evidence* where static features underdetermine identity. It returns only what a real interrogation would — one authoritative fact per call, not a full record.
2. **Virtual manipulation (verification)**. The agent *exercises a proposed correspondence*: it provisions a service through the candidate co-reference on a live side and confirms the resulting objects **in place against the semantic invariants** (§4). This is the paper's correctness-by-construction step, and it either **confirms** a proposal (invariants preserved) or **refutes** it (some invariant diverges). It is the only mechanism that can settle a structurally-symmetric pair from the graph itself.

The two are separated on purpose — interrogation feeds the *propose* stage, manipulation drives the *verify* stage (§7) — and their availability follows the spectrum:

- **both-cognitive** — both interrogation and manipulation answer about either side; a call budget (including *unbounded* as a limiting condition) is a variable.
- **one-inert** — the oracle serves only the live side; the inert side is static, so an experiment-only individual on the inert side is resolvable only if a published reference (§8) supplies its identity or its invariants.
- **both-inert** — the oracle is unavailable; static evidence only; the agent may only propose, and closure needs an external test or a human sign-off.

With an unbounded budget at both-cognitive the residual should vanish; constrain or remove the oracle and the experiment-only individuals reappear in the residual — the measurable form of "resolution complete in principle with full cognition."

7. The pipeline: propose → verify → refer, with confidence

Mirroring the schema study's proposal-then-verification-and-repair, and making the paper's "held with a confidence ... corroborated by regenerating native records" mechanical:

1. **Propose**. From the available static evidence (and interrogation, where allowed), the agent proposes co-reference pairs, **each with a confidence**, and marks the rest residual.
2. **Verify**. Each proposal is corroborated — minimally by regenerating a native record and checking fit, and, where cognition and budget allow, by a **virtual manipulation** against the invariants (§6). A proposal that survives is **confirmed**; one that is refuted is dropped (and its individuals returned to residual or re-proposed); one that can be neither confirmed nor refuted (no live access, underdetermined) is **referred onward**, not asserted.
3. **Refer**. The residual is reported and, crucially, **broken out** into native gap, experiment-only-unresolved, and low-confidence-unconfirmed — so the shortfall is attributable to its cause at each placement.

This makes the confirm-versus-propose distinction explicit, so recall is read exactly as in the schema study: a confirmed co-reference is asserted; an unconfirmed one is deferral, not error.

8. The reference family at the instance level

The reference is not only the key anchor. The paper names an **instance** variant and an **invariant** variant (among others), and both bear on this study, so the reference is an **axis**, not a fixed input:

- **none** — no shared reference; static evidence and live cognition only.
- **instance reference** — shared **opaque keys** plus canonical instance descriptors; the family analogue of the lexical reference, and the direct test of whether a published key substitutes for cognition (holding co-reference even where probing is gone).
- **invariant reference** — a published anchor that pins the **semantic invariants** for the individuals/services in play. This is what a virtual manipulation checks *against* when a side is inert and cannot itself be exercised: it lets an inert-side individual be tested against published invariants without a live experiment, so it is the mechanism expected to *partly* recover the experiment-only subset in the one-inert regime — bounded, because a reference can state the invariants but cannot run the experiment.

Running none / instance / invariant as arms is what tells us whether each reference variant does the work the paper

claims, and separates the key anchor's contribution from the invariant anchor's.

9. Metrics

Correctness is again the clean, length-independent currency for "which evidence and which mechanism matters":

- **Instance precision / recall** over the co-reference pairing, against the gold alignment — reported both at *proposal* and after *verification* (confirmed precision/recall), as in the schema verify-and-repair result.
- **Surviving instance false cognates** — distinct individuals wrongly merged (and, post-verify, how many the virtual manipulation caught).
- **Merge / split errors** — the entity-resolution failure modes (a single individual split across two records left unmerged; two individuals merged into one).
- **Invariant-confirmation rate** — of the true co-references, the fraction *confirmable* by virtual manipulation at each placement; the direct handle on resolution-complete-in-principle.
- **Confidence calibration** — proposed confidence against realised correctness, by model and placement.
- **Residual, broken out** — native gaps, experiment-only-unresolved, low-confidence-unconfirmed — the per-cause measure of the shortfall.
- **Effort** — **interrogation count** and **manipulation count** (the new task-specific currencies) reported alongside reasoning tokens.

10. Experimental design

The same shape as the schema studies: a **cognition-spectrum × evidence-ablation × reference-variant × model-ladder** cross over the seeded hard case, with a validated gold and the propose→verify→refer pipeline.

- **Placements**: both-cognitive, one-inert, both-inert (the spectrum is the primary axis).
- **Evidence ablation**: a factorial over {global key, name, attributes, topology} (provenance optional), the always-present anchor being the individual's own record identity within its graph. Full factorial if the run stays affordable, a fractional design if not.
- **Reference variant**: none / instance / invariant (§8).
- **Oracle budget**: at least {none, bounded, unbounded} at both-cognitive, to trace the shortfall against interrogation-and-manipulation cost.
- **Models**: the strength ladder (`gpt-5.6-sol` , `gpt-5-mini` , `gpt-5-nano`), for the capability interaction.
- **Trials**: enough for stable per-condition means given model variance (≥ 5).

The full cross is large; §on execution (to be settled) will fix which axes run at full factorial and which are sampled, and the checkpoint/resume and en-route-capture discipline is a build requirement from the start, as for the completed studies.

11. Pre-registered predictions

1. **Resolvability, sharper.** With an unbounded oracle at both-cognitive the residual $\rightarrow 0$; as cognition recedes and interrogation/manipulation are lost, the experiment-only individuals become unresolvable and the residual grows — a **larger** shortfall than in schema reconciliation, because for some individuals acting on the graph is the *only* route.
2. **The virtual manipulation is what closes the hard cases.** Structurally-symmetric pairs are resolved neither by static evidence nor by attribute interrogation but by the provision-and-read-back against invariants; their confirmed-recall tracks manipulation availability almost exclusively.
3. **Topology is the workhorse static feature**; its removal most increases merge/split errors, most at the inert placements where manipulation cannot compensate.
4. **Names are the trap surface** — without disambiguating features, shared local names drive wrong merges (instance false cognates).
5. **A shared key substitutes for cognition; the invariant reference substitutes for the experiment — partly.** A present opaque key holds co-reference even where the oracle is gone (the instance-family analogue of the lexical-reference result). The invariant reference recovers *some* experiment-only cases in the one-inert regime by letting an inert-side individual be checked against published invariants — but not all, since a reference can state invariants and cannot run the experiment.
6. **Confidence calibration degrades with capability** — weaker agents are more often confidently wrong, and lean on names and keys while under-exploiting topology, interrogation, and the virtual manipulation.

12. What it reuses, and what must be built

Reused as-is: the case/gold/derive-and-validate pattern, the cognition-spectrum treatment, the model ladder, the correctness-first measurement discipline, the proposal-then-verification shape of `verify_experiment.py`, and most metrics.

To build:

- an **instance-reconcile task** in the harness (propose co-reference pairs with confidence + residual over individuals, scored against an instance gold);
- the **oracle** on a live side, providing both **interrogation** (one authoritative fact per call) and the **virtual manipulation** (provision-and-read-back confirmed against the invariant set), with separate call counters; this is the main lift and worth building carefully, since it is what makes the spectrum meaningful for individuals and carries the paper's verification claim;
- a **verify-and-repair stage** for instances: corroborate each proposal by native-record regeneration and, where allowed, virtual manipulation; split the residual by cause;
- a **seeded instance case** over the reconciled TAPI/TEAS types, with the trap taxonomy of §4, the per-correspondence **invariant signatures**, and an `experiment-only` flag in the gold;
- the **instance and invariant references** (shared opaque keys / canonical instance descriptors; and a published invariant anchor) as the family analogues of the lexical reference, for the reference-variant axis and the with/without-reference contrast;
- checkpoint/resume with en-route row-by-row capture, from the start.

13. Open design questions

- **Oracle fidelity, per mechanism.** Interrogation returns one authoritative fact per call; the virtual manipulation returns the invariant check on the provisioned objects — no more than a real provisioning-and-read-back would. Both are bounded to what a real operation reveals; the fidelity ceiling is set so that easy cases are not made trivial and symmetric cases are not made impossible.
- **Separating "confirm" from "propose".** The verify stage must distinguish a proposed-but-unconfirmed co-reference (referred onward) from an asserted one, so recall is read the same way as in the schema study; §7 makes this mechanical.
- **Schema coupling.** This study assumes the schema is already reconciled. A later, harder variant would run schema and instance reconciliation *jointly*, where a tentative type alignment and a tentative instance alignment inform each other.
- **Invariant reference authoring.** How much the invariant reference publishes (which of the six invariants, at what granularity) is itself an ablatable question; the first run fixes a sensible full-invariant anchor and leaves the finer ablation to a follow-up.